



# **Investigation of Microbiome Abundances in Western vs. Non-Western Populations**

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|               |                               |
|---------------|-------------------------------|
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# Abstract

The human gut microbiome is shaped by diet, lifestyle, and geography, yet identifying microbial signatures that generalize across populations remains challenging due to high interpersonal variability and technical batch effects. This study investigates whether consistent microbial markers distinguish Western from Non-Western populations and characterize extreme BMI phenotypes, using metagenomic data from the Human Microbiome Atlas (3,292 healthy individuals, 20 countries, ~1,990 MSPs).

After CLR transformation and diagnostic validation for batch and library-size effects, unsupervised ordination (UMAP, t-SNE) revealed a separation between Western and Non-Western lifestyles but no comparable structure for BMI within the USA cohort. A supervised pipeline combining Random Forest and L1-regularised Logistic Regression, validated through nested cross-validation with Stratified Group K-Fold partitioning by country, was therefore applied.

Both classifiers converged on five shared Western/Non-Western drivers with statistically significant medium-to-large effect sizes (Random Forest ROC-AUC  $\approx$  0.95): *Ruminococcaceae* (Non-Western) and *Eubacterium ramulus*, *Anaerostipes*, *Anaeromassilibacillus*, and *Blautia* (Western). For BMI extremes in the USA cohort, a label-permutation test confirmed a subtle but valid signal (balanced accuracy 0.570,  $p = 0.02$ ), with *Bacteroides uniformis* and two *Alistipes* species enriched in lean individuals, and *Clostridium* sp. CAG:169 and *Blautia* sp. SG772 enriched in obese individuals. While several findings align with existing literature, others (notably the Western enrichment of *Anaerostipes* and *Blautia*, and the lean-enrichment of *A. onderdonkii*) diverge from expected patterns, highlighting the context-dependent and strain-specific nature of gut–host associations.

## Abstrakt (Deutsch)

Das humane Darmmikrobiom wird durch Ernährung, Lebensstil und Geografie geprägt. Die Identifizierung mikrobieller Signaturen, die sich über verschiedene Populationen hinweg generalisieren lassen, bleibt jedoch aufgrund der hohen interindividuellen Variabilität und technischer Batch-Effekte eine Herausforderung. Diese Studie untersucht anhand von metagenomischen Daten aus dem Human Microbiome Atlas (3.292 gesunde Individuen, 20 Länder, ~1.990 MSPs), ob konsistente mikrobielle Marker westliche von nicht-westlichen Populationen unterscheiden und extreme BMI-Phänotypen charakterisieren.

Nach CLR-Transformation und diagnostischer Validierung bezüglich Batch- und Library-Size-Effekten zeigte die unüberwachte Ordination (UMAP, t-SNE) eine Trennung zwischen westlichen und nicht-westlichen Lebensstilen, jedoch keine vergleichbare Struktur für den BMI innerhalb der USA-Kohorte. Daher wurde eine überwachte Pipeline angewendet, die Random Forest und L1-regularisierte logistische Regression kombiniert und mittels geschachtelter Kreuzvalidierung (Nested Cross-Validation) mit landesspezifischer Stratified Group K-Fold-Partitionierung validiert wurde.

Beide Klassifikatoren konvergierten auf fünf gemeinsame westliche/nicht-westliche Treiber mit statistisch signifikanten, mittleren bis großen Effektstärken (Random Forest ROC-AUC  $\approx$  0,95): *Ruminococcaceae* (nicht-westlich) sowie *Eubacterium ramulus*, *Anaerostipes*, *Anaeromassilibacillus* und *Blautia* (westlich). Für die BMI-Extreme in der USA-Kohorte bestätigte ein Label-Permutationstest ein dezentes, aber valides Signal (balancierte Genauigkeit 0,570,  $p = 0,02$ ). Dabei zeigten sich *Bacteroides uniformis* und zwei *Alistipes*-Arten bei schlanken Individuen angereichert, während *Clostridium* sp. CAG:169 und *Blautia* sp. SG772 bei fettleibigen Individuen angereichert waren. Während einige Befunde mit der bestehenden Literatur übereinstimmen, weichen andere (insbesondere die westliche Anreicherung von *Anaerostipes* und *Blautia* sowie die Anreicherung von *A. onderdonkii* bei schlanken Individuen) von erwarteten Mustern ab. Dies unterstreicht die kontextabhängige und stammspezifische Natur von Assoziationen zwischen Mikrobiom und Wirt. [1]

# Acknowledgement

We would like to express our deepest gratitude to our supervisor, Dr. Tugçe Bilgin Sonay, for her invaluable expertise, constant guidance, and support throughout this project. Her mentorship not only allowed us to explore an exciting area of microbiome research but also significantly advanced our technical skillset, academic understanding, and scientific curiosity.

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# List of Abbreviations

*The following abbreviations are used throughout this project work on the investigation of microbiome abundances in Western vs. Non-Western populations.*

| Abbreviation  | Meaning                                                     |
|---------------|-------------------------------------------------------------|
| <b>5-FU</b>   | 5-Fluorouracil (chemotherapeutic agent)                     |
| <b>AI</b>     | Artificial Intelligence                                     |
| <b>AUC</b>    | Area Under the Curve                                        |
| <b>BMI</b>    | Body Mass Index                                             |
| <b>CAG</b>    | Co-Abundance Gene group                                     |
| <b>CLR</b>    | Centered Log-Ratio (transformation)                         |
| <b>ComBat</b> | Combating Batch Effects (Empirical Bayes correction method) |
| <b>CV</b>     | Cross-Validation                                            |
| <b>DNA</b>    | Deoxyribonucleic Acid                                       |
| <b>DRS</b>    | DDBJ Read Submission (SRA accession prefix)                 |
| <b>ERR</b>    | European Run Read (SRA accession prefix)                    |
| <b>F1</b>     | F1 Score (harmonic mean of precision and recall)            |
| <b>GEM</b>    | Genome-scale Metabolic Model                                |
| <b>HDL</b>    | High-Density Lipoprotein                                    |
| <b>HMA</b>    | Human Microbiome Atlas                                      |
| <b>L1</b>     | L1 Regularization (LASSO penalty)                           |
| <b>LASSO</b>  | Least Absolute Shrinkage and Selection Operator             |
| <b>LR</b>     | Logistic Regression                                         |
| <b>MDI</b>    | Mean Decrease in Impurity (Gini importance)                 |

| Abbreviation   | Meaning                                                  |
|----------------|----------------------------------------------------------|
| <b>ML</b>      | Machine Learning                                         |
| <b>MSP</b>     | Metagenomic Species Pangenome                            |
| <b>NAFLD</b>   | Non-Alcoholic Fatty Liver Disease                        |
| <b>PC1</b>     | First Principal Component                                |
| <b>PC2</b>     | Second Principal Component                               |
| <b>PCA</b>     | Principal Component Analysis                             |
| <b>R</b>       | Pearson Correlation Coefficient / R programming language |
| <b>RF</b>      | Random Forest                                            |
| <b>ROC</b>     | Receiver Operating Characteristic                        |
| <b>ROC-AUC</b> | Area Under the Receiver Operating Characteristic Curve   |
| <b>SCFA</b>    | Short-Chain Fatty Acid                                   |
| <b>SRA</b>     | Sequence Read Archive                                    |
| <b>SRR</b>     | Sequence Read Run (SRA accession prefix)                 |
| <b>TSS</b>     | Total Sum Scaling                                        |
| <b>t-SNE</b>   | t-distributed Stochastic Neighbour Embedding             |
| <b>UK</b>      | United Kingdom                                           |
| <b>UMAP</b>    | Uniform Manifold Approximation and Projection            |
| <b>USA</b>     | United States of America                                 |

# 1. Introduction

## 1.1. The Gut Microbiome as a Metabolic Organ

The human gut microbiome is currently defined as an essential, dynamic "metabolic organ" that plays a pivotal role in human health [2]. It serves as a biological fingerprint, primarily shaped by host genetics and external factors such as diet and lifestyle [3]. These microbial communities are not only responsible for digestion but also engage in complex interactions with the immune system and metabolism, where high diversity is generally considered an indicator of a robust state of health [2].

As global urbanization continues to shift traditional lifestyles toward "Westernization", characterized by highly processed foods and altered hygiene standards, a significant change in microbial composition is observable [2]. These shifts, often termed dysbiosis, are closely linked to the rise of non-communicable diseases, including obesity, type 2 diabetes, and chronic inflammatory processes [2].

## 1.2. Research Gap: Identifying Generalizable Microbial Signatures

While it is well-established that geography and diet influence the human gut, identifying microbial signatures that remain consistent across diverse global populations remains a significant challenge [4]. Even in healthy individuals, the abundance of shared taxa can vary by more than an order of magnitude [4], masking patterns that distinguish one cultural group from another. Technical "batch effects" between country-specific datasets further blur whether a signature reflects a true biological trait or differing sampling methodologies. A streamlined analytical approach is therefore required to identify bacterial groups characteristic of specific lifestyles, regardless of country of origin.

Despite mechanistic advances linking specific microbial taxa to obesity, type 2 diabetes, and inflammatory bowel disease, the field still lacks robust, population-generalizable signatures, requiring analytical pipelines capable of separating biological signal from technical noise in high-dimensional gut data [3], [4].

### 1.3. Study Objectives and Analytical Approach

The primary objective of this project is to identify and compare microbial differences between different countries to characterize global variation. By utilizing machine learning models, the work seeks to analyse microbial signatures and their functional relevance to the host. The project is guided by the following research goals:

- **Signature Analysis:** To determine whether specific microbial markers can be identified that are characteristic of different groups or populations.
- **Bacteria Characterization:** To determine the biological functions of specific bacteria and establish their link to different physiological traits.
- **Clinical Correlation:** To evaluate whether the identified microbial markers align with established clinical literature on metabolic disease, including obesity, type 2 diabetes, and cardiovascular risk profiles.

## 2. Materials and Methodology

### 2.1. Data Source and Study

The analysis was implemented primarily in Python with selected steps in R; all Jupyter Notebooks are publicly available on GitHub [5].

#### 2.1.1. The Human Microbiome Atlas (HMA)

The foundation of this project is the Human Microbiome Atlas (HMA) [6], a comprehensive resource that integrates global metagenomic data with high-quality genome-scale metabolic models (GEMs). This dataset allows for a high-resolution analysis of the gut microbiome across diverse populations and health status, providing both taxonomic and potential functional insights.

#### 2.1.2. Data Streams and Primary Components

This project utilizes three integrated data streams from the HMA to bridge the gap between genetic presence and biological context:

1. **Abundance Matrix (vect\_atlas.csv):** This core file contains the microbial profiles. Each row represents a Metagenomic Species Pangenome (MSP), and each column represents an individual sample identified by a unique SRA Accession Number (prefixes DRS, ERR, or SRR). The numerical values represent the relative abundance of each taxon within a sample.
2. **Metadata (sample\_id table):** This provides the demographic and technical context for each sample. Key variables include age, gender, BMI, geography, Bio project, and technical sequencing metrics such as Gene richness and Mgs Richness (alpha diversity metric).
3. **Metabolic Models (MSPmodels):** For each MSP identified in the abundance matrix, a corresponding metabolic model is available in .mat format. These models contain the metabolic "blueprints" of the bacteria, enabling future functional inference beyond simple taxonomic identification.

#### 2.1.3. Cohort Consolidation and Inclusion Criteria

The first step in the exploration phase involved merging the abundance matrix with the metadata table. To ensure the baseline analysis reflected a "normal" physiological state, an inclusion filter was applied:

- **Health Status:** Only samples labelled as **healthy column** were retained. This was done to ensure that subsequent comparisons were not confounded by clinical pathologies or acute infections.

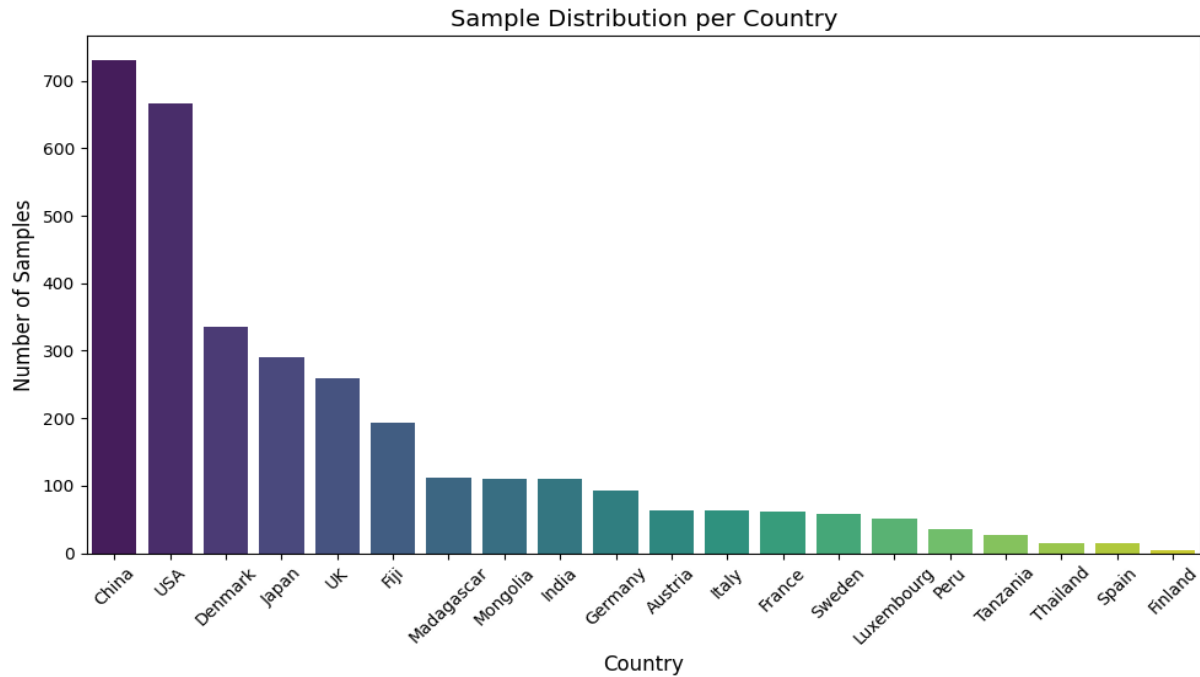
#### 2.1.4. Cohort Characteristics and Global Distribution

After merging the taxonomic data with the demographic information and applying the health filter, the resulting consolidated data frame reached following dimensions:

- **Observations (3292 rows):** Each row represents a unique healthy individual from the global dataset.
- **Features (2004 columns):** The columns represent a high-dimensional feature space consisting of:
  - **Metagenomic Features:** Approximately 1,990 specific MSPs, capturing the diverse microbial landscape of the human gut.
  - **Demographic/Technical Features:** The 8 variables from the metadata *Sample ID, BMI, Geography, Gender, Age, Bioproject, MgsRichness, Gene Richness*, which provide the necessary context for subsequent supervised learning tasks.

This high-dimensional structure is a characteristic of metagenomic "Big Data." The extreme sparsity, evidenced by the high frequency of zero-values for rare MSPs, presents a significant computational challenge that necessitates the specialized preprocessing and normalization pipelines described in the following chapter.

## 2.2. Data Preprocessing and Quality Control



*Figure 1: Global distribution of the 3,292 healthy samples across the 20 countries included in the cohort. Bar height represents the number of individuals per country, ordered by sample size*

The study cohort is distributed across 20 countries, though the representation varies significantly (Figure 1). The dataset is primarily driven by large cohorts from China ( $n = 730$ ), the USA ( $n = 666$ ), and Denmark ( $n = 335$ ), which together account for over half of the total observations. Other regions, such as Madagascar ( $n = 112$ ), Mongolia ( $n = 110$ ) and Tanzania ( $n = 27$ ) provide smaller but unique geographical subsets. The smallest cohort are Spain ( $n = 14$ ) and Finland ( $n = 4$ ). This dataset is dominated by a few large-scale studies, creating a significant geographical imbalance.

Beyond geography, it is essential to consider how samples are distributed across individual studies, identified by their Bio Project IDs (Figure 2). The dataset comprises a mixture of large-scale studies spanning multiple countries and smaller, more localized cohorts.

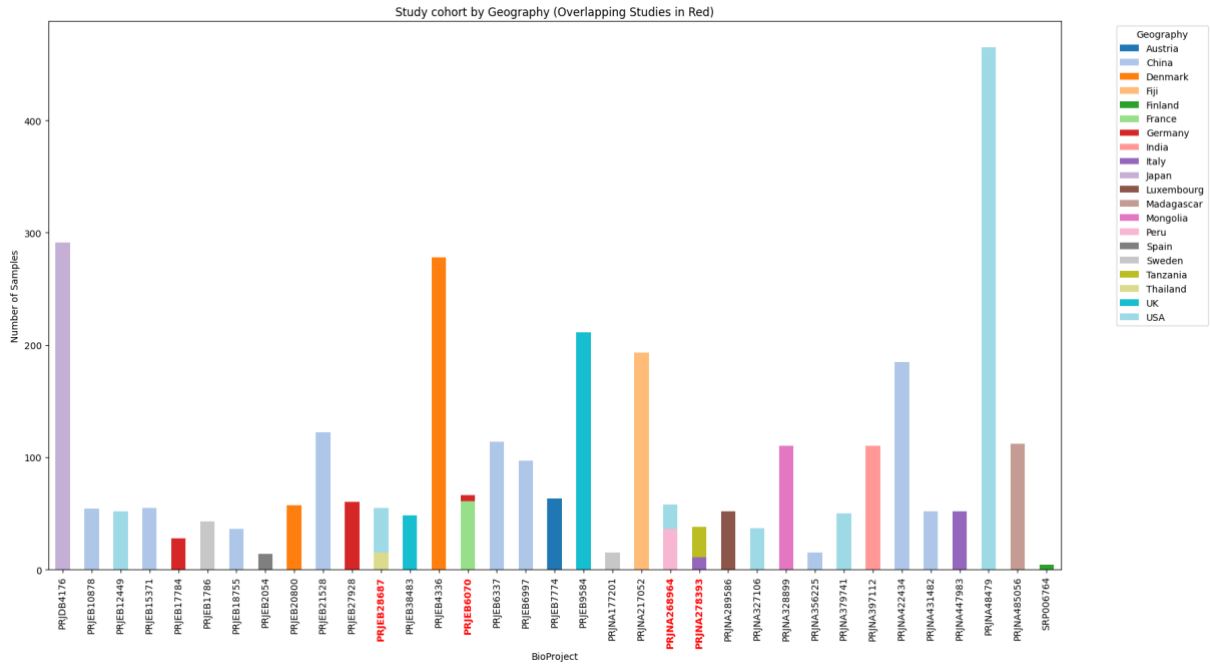


Figure 2: Sample counts per BioProject. Each bar represents one BioProject, with bar height indicating the number of samples and bar colour indicating the country of origin. BioProject IDs highlighted in red on the x-axis denote studies that contribute samples to more than one country, illustrating the overlap between study identity and geography

This uneven distribution is a primary source of potential batch effects. These technical variations, resulting from differences in DNA extraction, library preparation, or sequencing protocols, can introduce "noise" into the data. Without careful inspection, machine learning algorithms may inadvertently split the data based on study-specific biases rather than true biological signatures.

### 2.2.1. Workflow

Before applying analytical methods or machine learning pipelines, raw metagenomic data must undergo rigorous preprocessing to account for its unique mathematical properties. Bacterial abundance tables are inherently sparse and compositional [7], with high zero-inflation. This sparsity arises because individual samples only contain a specific subset of the global microbial diversity, and limited sequencing depth, often referred to as sampling effort, can lead to a failure in capturing rare taxa [8]. To ensure a robust comparison between geographical cohorts and phenotypes, the following pipeline was implemented based on established best practices [7], [8].

### 2.2.2. Quality and Prevalence Filtering

Filtering was performed prior to normalization to control for technical variation and sequencing errors.

- **Sample Filtering (Richness):** To exclude samples with insufficient sampling effort, we removed those in the lowest 5th percentile of richness. This ensures that extremely low richness, which typically indicates technical artifacts or poor DNA yield, does not bias the analysis [8].
- **Taxa Filtering (Prevalence):** MSPs present in fewer than 10% of samples were removed. This reduces sparsity and noise, as rare sequence diversity often arises from sequencing errors. Focusing on stable microbial signatures leads to more reproducible ML insights [8].

### 2.2.3. Normalization: Total Sum Scaling (TSS)

To account for varying sequencing depths, counts were converted to relative abundances by dividing each sample by its row sum:

$$x_{rel} = \frac{x_i}{\sum x_i}$$

While TSS makes samples comparable, it reinforces the compositional nature of the data. In this closed system, an increase in one species forces others to decrease mathematically, which can trigger wrong correlations [7].

### 2.2.4. Transformation: Centred Log-Ratio (CLR)

To break the dependency of compositional data and move into a real-number space where standard statistical tools (like Random Forest) are valid, we applied the CLR transformation. It expresses each taxon relative to the geometric mean of the sample:

$$CLR(x) = \left[ \ln \frac{x_1}{g(x)}, \dots, \ln \frac{x_n}{g(x)} \right]$$

$$g(x) = \sqrt[n]{x_1 \cdot x_2 \dots x_n}$$

Where  $g(x)$  is the geometric mean of all taxon abundances within the sample. Zeroes were replaced by a pseudo-count of  $1 \times 10^{-6}$ .

### 2.2.5. Visual Validation: CLR Distribution

The following plot illustrates the distribution of values after the CLR transformation. This visualization is crucial to confirm that the transformation has successfully handled the high sparsity of the raw data.

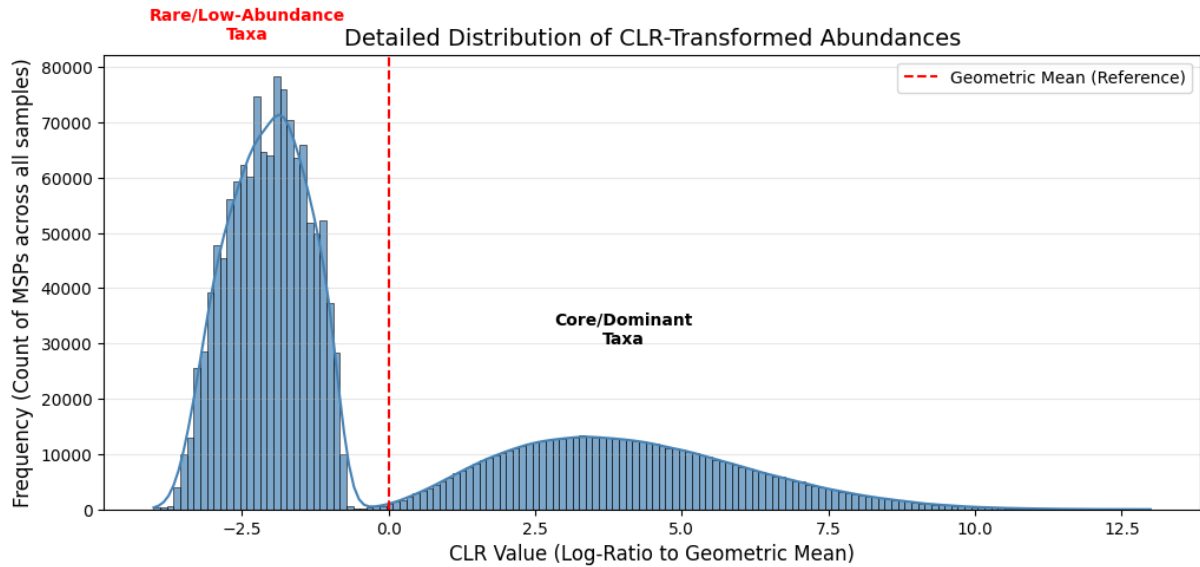


Figure 3: Density distribution of MSP abundance values after centred log-ratio (CLR) transformation, aggregated across all samples. The red dashed line marks the per-sample geometric mean, which serves as the internal reference of the CLR transformation

The transformed distribution in Figure 3 is bimodal: the left peak corresponds to the "rare biosphere" (low-abundance taxa, including pseudo-count-imputed zeros) and the right peak to the dominant taxa of each sample. The red dashed line marks the per-sample geometric mean used as the internal reference. The resulting symmetric, log-ratio distribution is suitable for downstream machine learning.

### 2.2.6. Batch Effect Assessment

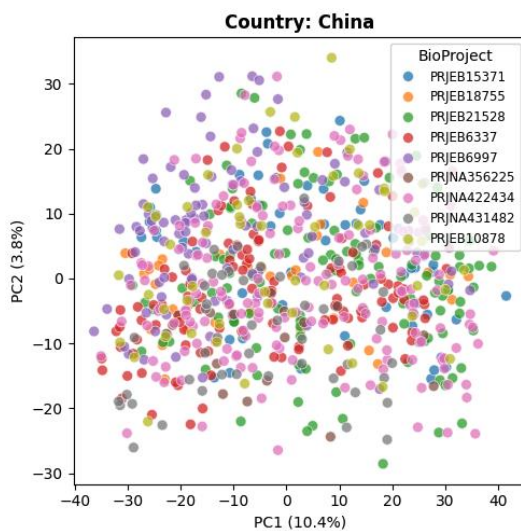
Batch effects arise when samples are processed across different laboratories, technologies, protocols, operators, or time points [9]. Such technical variation must be corrected prior to downstream analysis to ensure sample comparability.

Among available correction methods, the Empirical Bayes approach (implemented in ComBat) performs robustly on high-dimensional data with small sample sizes, making it well-suited to microbiome abundance data. By leveraging information across all samples to estimate batch-associated variation, it yields reliable corrections even under limited sample conditions. To reduce noise, taxa absent in more than 80% of samples were filtered out prior to correction, following established recommendations [10].

A central challenge is the structural overlap between technical and biological variation: each country is largely represented by a distinct cohort, confounding country differences with

study-related batch effects. A naive correction could therefore remove biologically meaningful signal alongside technical artefacts. To assess this, a principal component analysis (PCA) was performed separately for each country represented by more than one cohort, allowing batch effects to be evaluated independently of inter-country differences.

Across all examined countries (China, Denmark, Germany, Italy, Sweden, UK, USA), samples from different BioProjects largely overlap in the first two principal components, with no clear clustering by cohort. This can be seen in Figure 4 as an example for China. PC1 and PC2 explain only a modest share of the variance (PC1: 8.2–10.6%; PC2: 3.8–7.4%), and BioProject distributions appear diffuse rather than structured. Minor visual offsets in Denmark do not form distinct clusters (see Appendix [Figure 13](#) for all country specific PCA plots).



*Figure 4: Principal component analysis (PC1 vs. PC2) of CLR-transformed microbiome profiles for the Chinese cohort, coloured by BioProject. PC1 and PC2 explain 10.4% and 3.8% of the total variance, respectively. Shown as a representative example of the per-country batch-effect assessment performed for all countries with multiple BioProjects (full set in Appendix Figure 15)*

### 2.2.7. Library Effect Assessment

Although the centred log-ratio (CLR) transformation provides a robust framework for handling the compositional nature of microbiome data, it can remain sensitive to differences in library size (the total bacterial counts per sample), particularly in sparse datasets typical of microbiome studies. If unaddressed, dominant patterns may reflect sequencing depth rather than genuine biological signal, increasing the risk of Type I errors in cohort comparisons [11].

To assess residual library size effects, a Log-Ratio PCA diagnostic was performed. The Pearson correlation ( $R$ ) between  $\log_{10}$ -transformed library sizes and the first principal component (PC1) served as a quantitative measure of association. A low correlation combined with a non-linear, diffuse sample distribution was defined as the success criterion, indicating effective decoupling of biological signal from sequencing depth [11].

The diagnostic plot ((see Appendix Figure 14) yielded  $R = 0.160$ , indicating a weak association between sequencing depth and the dominant axis of biological variation. Samples form a diffuse, non-linear cloud rather than a directional trend, with PC1 values spanning a comparable range across all sequencing depths. These results suggest that PC1 captures genuine biological variation and that the CLR transformation has effectively decoupled the biological signal from library size. No additional normalisation was therefore deemed necessary.

## 2.3. Summary of Preprocessing

Quality filtering, TSS normalisation and CLR transformation, together with the batch- and library-size diagnostics above, yielded a symmetric log-ratio distribution free of dominant technical artefacts. No additional correction was applied, and the resulting matrix forms the input for the unsupervised and supervised analyses that follow.

## 2.4. Unsupervised Learning

After preprocessing, CLR transformations, batch effect and library effect testing, the dataset remains high-dimensional. To extract meaningful biological signals from sparse, compositional microbiome data, we employ non-linear dimensionality reduction.

### 2.4.1. Methodological Rationale

Conventional linear PCA is often discouraged for microbiome data, as it assumes linear relationships and can mask the complex, non-linear structure typical of compositional gut profiles [12], [13]. We therefore apply t-SNE and UMAP directly to the CLR-transformed matrix: t-SNE preserves local neighbour structure by converting distances into conditional probabilities, while UMAP uses topological manifold learning to balance local clusters against broader global structure [12]. Distances between samples are computed using the Aitchison distance, which is mathematically equivalent to the Euclidean distance on CLR-transformed data [13]. Because it operates on log-ratios rather than absolute counts, it depends only on the internal balance of the microbial community and is unaffected by sequencing depth or the arbitrary scale of relative abundances.

### 2.4.2. Global Exploratory Ordination

To identify the primary drivers of microbial variance, we first visualized the entire dataset using UMAP. This "World View" helps determine whether geographical origin alone is enough to form distinct biological clusters.

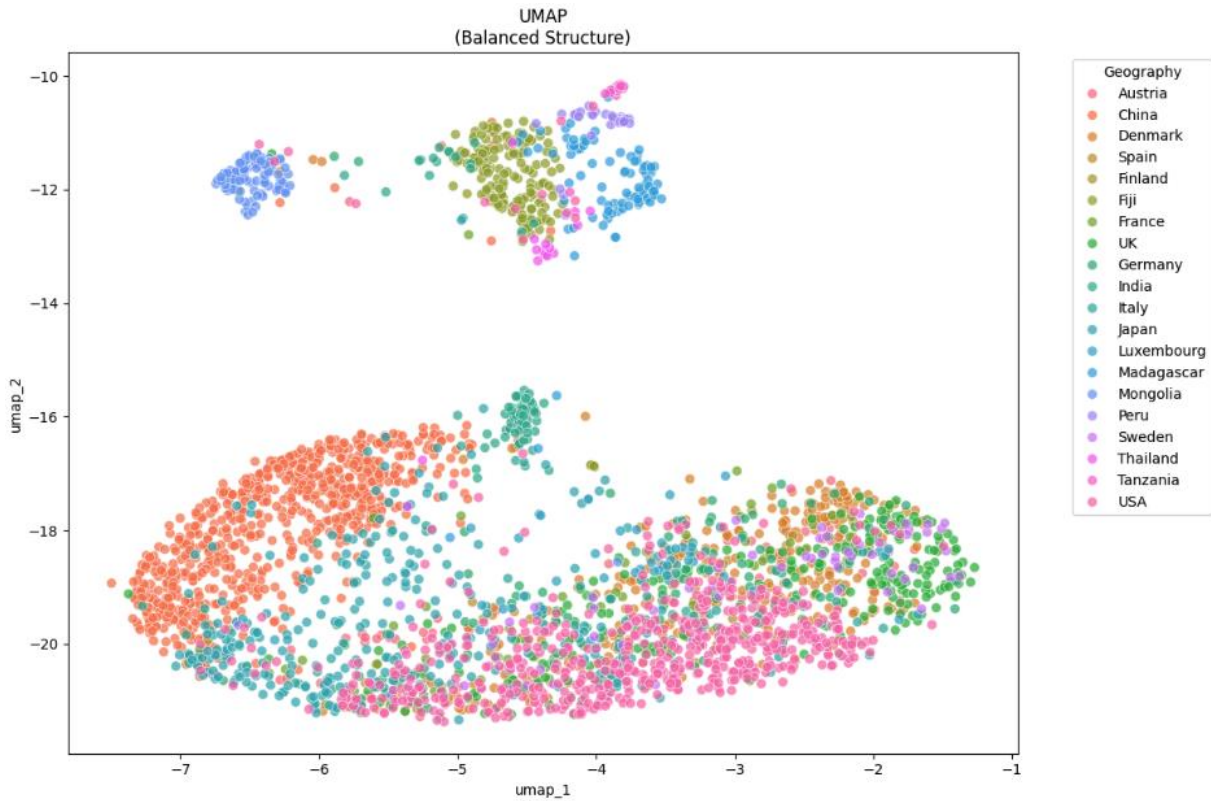


Figure 5: UMAP ordination of all healthy samples based on Aitchison distances of CLR-transformed MSP abundances. Each point represents one individual, coloured by country of origin.

Initially, mapping the data by individual nations showed clear differences, indicating that location strongly influences microbes. However, with 20 different countries, the visualization lacked resolution, especially in the bottom cluster. Grouping the samples by specific categories will significantly improve data readability.

### 2.4.3. Western vs Non-Western

The western group contains: Austria, Denmark, France, Germany, Italy, Luxembourg, Spain, Sweden, United Kingdom, USA.

The Non-western group contains: China, India, Japan, Mongolia, Thailand, Madagascar, Tanzania, Peru, Fiji.

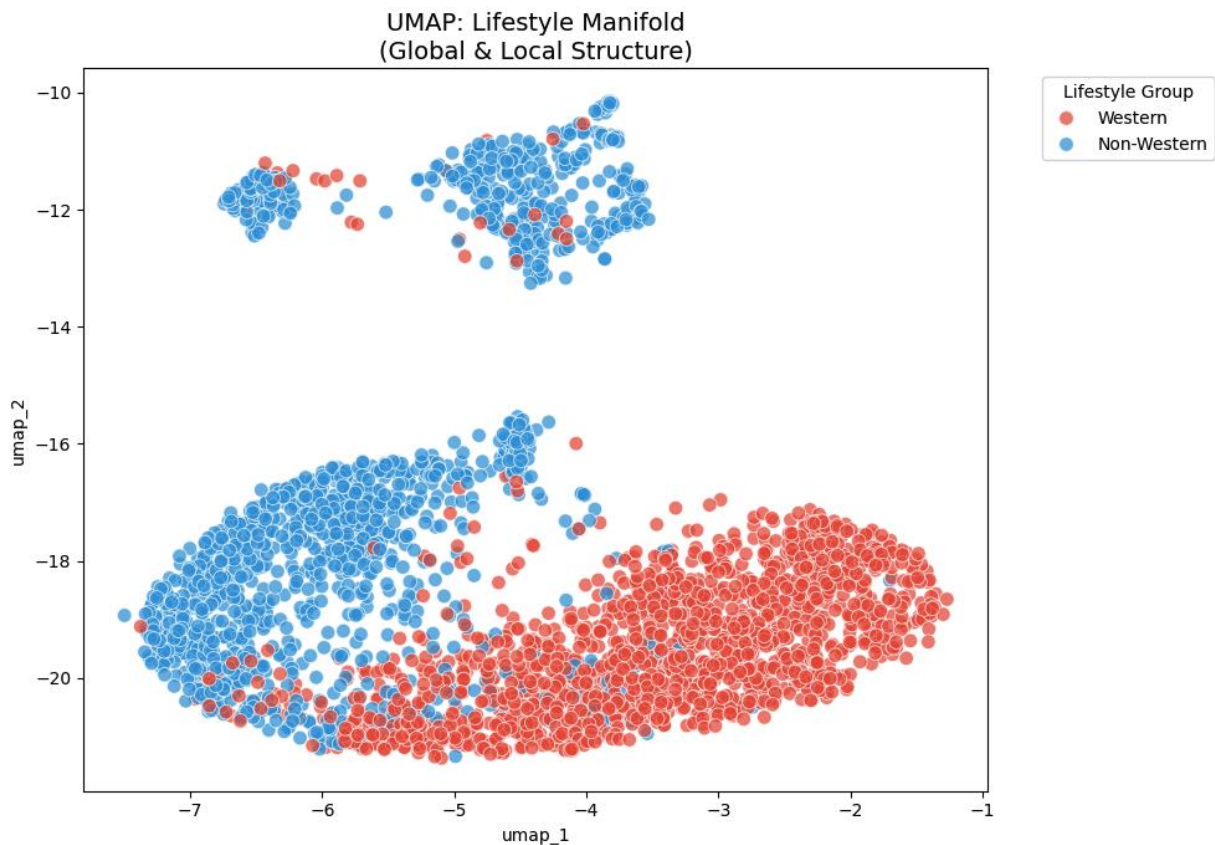


Figure 6: UMAP ordination of the same dataset as in Figure 5, re-coloured by lifestyle group. Western: Austria, Denmark, France, Germany, Italy, Luxembourg, Spain, Sweden, UK, USA; Non-Western: China, India, Japan, Mongolia, Thailand, Madagascar, Tanzania, Peru, Fiji

UMAP by Lifestyle group shows significant visual separation, confirming that industrialization and socio-ecological factors play an important role in microbial variance. The Human Microbiome Atlas treated the USA, China, and Japan as distinct cohorts; we assigned them to "Western" and "Non-Western", respectively. This simplifies the analysis and avoids the unnecessary complexity of treating them as separate groups.

#### 2.4.4. Exploratory screening of dietary and socioeconomic variables

In addition to lifestyle, we explored several other potential drivers of microbial variance, including meat consumption [14], life expectancy [15], sugar consumption [16], lactase persistence [17], and national development status [18].

While initial visualizations showed varying degrees of clustering, these patterns were found to be heavily confounded by geographical origin. For example, clusters for "high meat consumption" overlapped almost entirely with specific Western national cohorts. Because these variables did not provide an independent biological signal beyond what was already

captured by the Country or Lifestyle labels, they were excluded from the final supervised learning pipeline to prevent redundancy and over-fitting.

### 2.4.5. BMI in the USA cohort

The final exploratory ordination focused on the phenotypic target: Body Mass Index (BMI). To ensure the highest data quality and minimize geographical confounding, this analysis was restricted to the USA cohort ( $n = 666$ ).

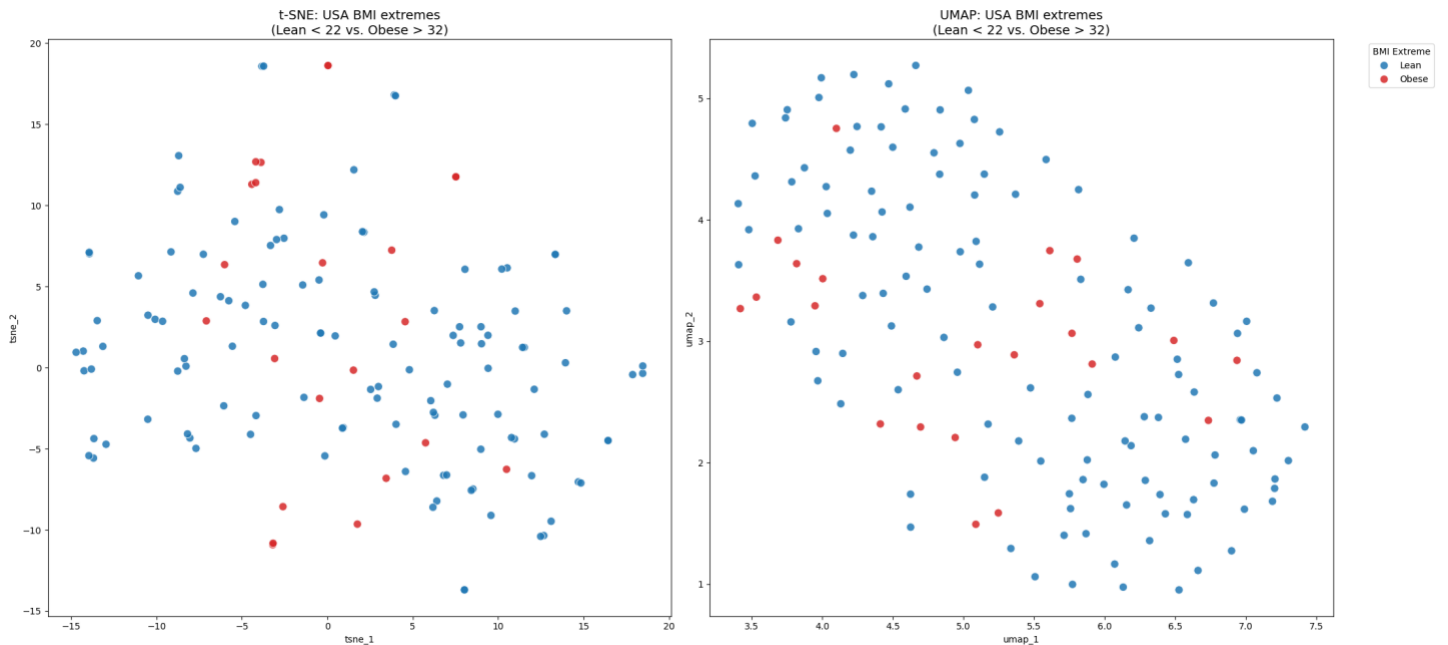


Figure 7: t-SNE (left) and UMAP (right) projections of the USA cohort ( $n = 666$ ) restricted to BMI extremes. Samples are coloured by group: Lean (BMI < 22,  $n = 119$ ) and Obese (BMI > 32,  $n = 23$ ).

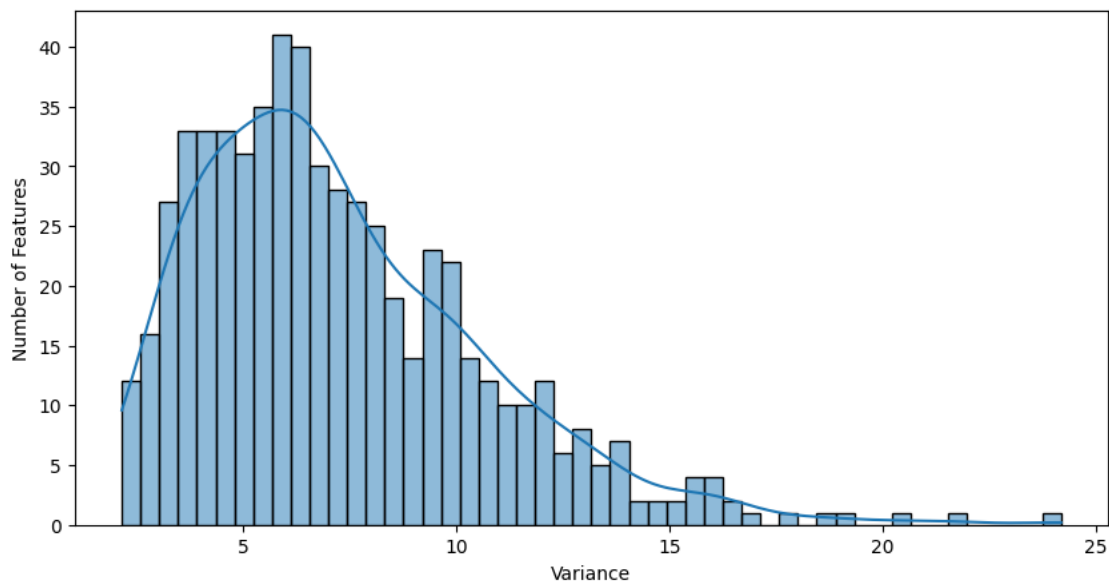
Initially, the cohort was visualized across three categories: Normal, Overweight, and Obese. However, to maximize the potential for detecting a biological signal, we followed a "tail-comparison" strategy, focusing exclusively on the extremes: Lean (BMI < 22,  $n = 119$ ) versus Obese (BMI > 32,  $n = 23$ ).

Despite this optimized contrast, the UMAP reveals no distinct visual clustering, with samples from both groups remaining entirely interspersed. This lack of visual partitioning suggests that BMI does not act as a dominant "macro-driver" of the total microbial community structure in the same way as lifestyle or geography. Instead, any microbial signature associated with adiposity is likely subtle and high-dimensional, providing the primary rationale for the use of supervised learning methods in the following chapters.

## 2.5. Supervised Learning: Methodology and Preliminary Performance

### 2.5.1. Pipeline Design and Feature Selection Strategy

Prior to model construction, an embedded feature selection strategy was adopted, in which selection occurs concurrently with model training. This approach is particularly well suited to high-dimensional data. A Random Forest classifier and a logistic regression model were implemented in parallel, as combining linear and non-linear approaches has been shown to improve predictive accuracy on complex biological datasets [19]. As an initial filtering step, near-zero-variance features were removed, since taxa with minimal variation across samples contribute negligible discriminative information. The distribution of feature variances is shown in Figure 8.



*Figure 8: Distribution of per-feature variances across CLR-transformed MSPs retained after the near-zero-variance filter. Bars show the histogram of feature counts, with a kernel density estimate overlaid. The distribution is right-skewed*

LASSO regression was additionally applied as a complementary feature selection method, using its L1 penalty to eliminate uninformative features. Model performance was assessed via nested cross-validation, with an inner loop for hyperparameter tuning and an outer loop providing an unbiased generalisation estimate. Stratified Group K-Fold partitioning was further employed to mitigate batch effects, assigning all samples from a given country exclusively to either training or test folds and thereby ensuring that performance reflects a transferable biological signal rather than cohort-specific artefacts [19].

## 2.5.2. Western vs. Non-Western Classification

### Random Forest

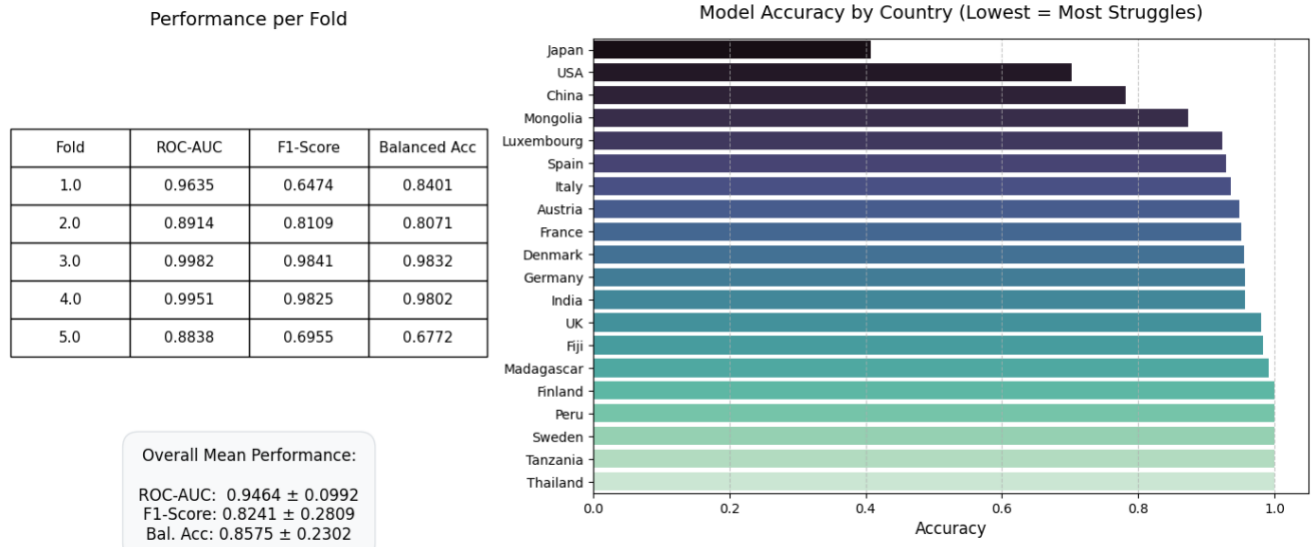


Figure 9: Performance of the Random Forest classifier for Western vs. Non-Western classification, evaluated by nested cross-validation with Stratified Group K-Fold partitioning. Shown are global metrics (ROC-AUC, balanced accuracy) and per-country balanced accuracy.

The Random Forest achieved a mean ROC-AUC of  $\approx 0.95$  and a mean balanced accuracy of 83%, but with substantial fold-to-fold variability ( $\pm 0.22$ – $0.27$ ). Per-country results (Figure 9) show that this instability is driven by a few cohorts: classification of Japanese samples failed almost entirely ( $\approx 40\%$ , worse than chance), with the United States ( $\approx 70\%$ ) and China ( $\approx 78\%$ ) also below the global mean. Overall performance is therefore high but does not generalise equally across populations. Randomised Search within the nested CV consistently selected the same configuration (variance threshold 0.0001, 200 trees, no depth limit, LASSO  $C = 1.0$ ). The final model was refit on the full dataset, and feature importances were extracted from the Random Forest stage for species surviving both the variance and LASSO filters.

Different classifications were analysed, as seen in Table 1.

Table 1: Comparing different classification strategies for the Random Forest

| Metric    | Western vs non-Western | Western vs non-Western vs Japan | Western vs non-Western without Japan | Western vs non-Western vs Transitioning | Western vs non-Western vs Transitioning vs Japan | Western vs non-Western vs US/China without Japan |
|-----------|------------------------|---------------------------------|--------------------------------------|-----------------------------------------|--------------------------------------------------|--------------------------------------------------|
|           | StratifiedKFold        |                                 |                                      | StratifiedKFold                         |                                                  |                                                  |
| ROC-AUC   | $0.9454 \pm 0.1015$    | $0.9881 \pm 0.0073$             | $0.9633 \pm 0.0535$                  | $0.9193 \pm 0.1113$                     | $0.9906 \pm 0.0028$                              | $0.8658 \pm 0.0769$                              |
| F1-Score  | $0.8196 \pm 0.2022$    | $0.8233 \pm 0.0436$             | $0.7215 \pm 0.3405$                  | $0.7050 \pm 0.2477$                     | $0.8592 \pm 0.0142$                              | $0.6757 \pm 0.1015$                              |
| Bal. Acc. | $0.8510 \pm 0.2411$    | $0.7863 \pm 0.0371$             | $0.7948 \pm 0.3859$                  | $0.7452 \pm 0.1863$                     | $0.8326 \pm 0.0150$                              | $0.6988 \pm 0.1280$                              |

Among the classification setups evaluated, the original binary separation into "Western" and "Non-Western" offers the most favourable trade-off between predictive performance and biological interpretability and was therefore retained for downstream analysis.

## Logistic Regression

The L1-regularised Logistic Regression provides a linear counterpart: its sparse, signed coefficients allow each top microbial driver to be independently validated and unambiguously assigned to the Western or Non-Western group based on the sign and magnitude of its estimated effect. The performance can be seen in Figure 10.

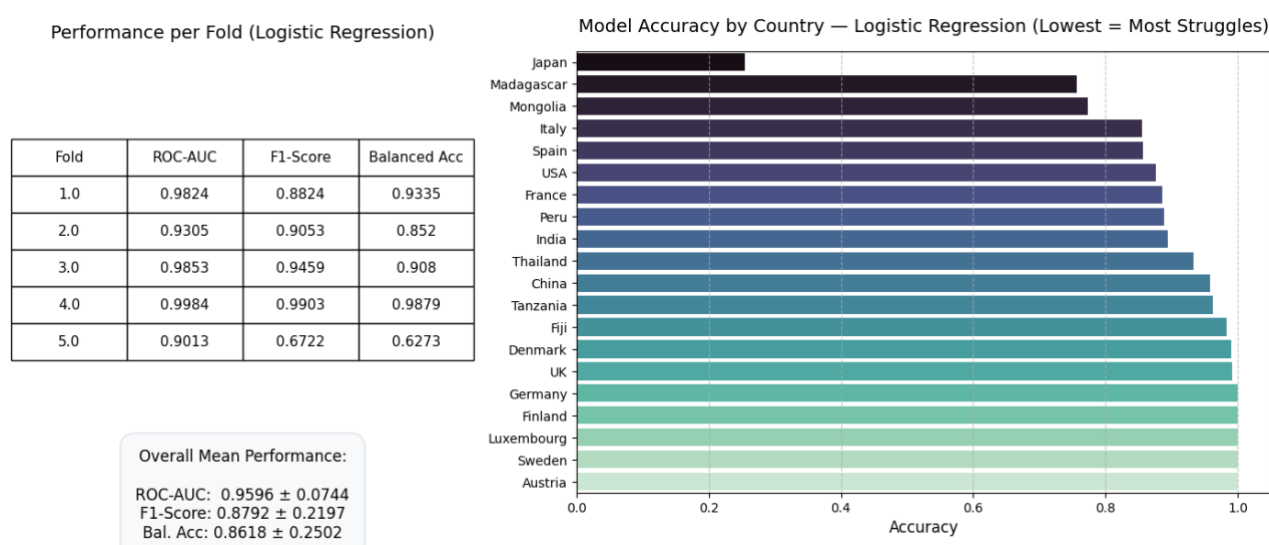


Figure 10: Performance of the Logistic Regression classifier for Western vs. Non-Western classification. Shown are global metrics (ROC-AUC, balanced accuracy) and per-country balanced accuracy.

After validating both models, each classifier was refit on the entire dataset to extract the microbial species that contributed most strongly to the Western vs. Non-Western decision boundary. The two models provide complementary perspectives: Logistic Regression yields directly interpretable linear coefficients, whereas Random Forest requires post-hoc explanation through feature importances combined with partial dependence analysis. In both cases, each top-ranked species was assigned a directional label ("Western" or "Non-Western") based on the sign of its contribution.

### 2.5.3. Statistical Analysis of microbiome compared in Western/Non-Western

To validate the differences between Western and Non-Western samples, we follow the approach of Gupta et al. (2020), who combined a Mann–Whitney U test with Cliff's delta to identify microbial species separating healthy from non-healthy guts [20].

The Mann–Whitney U test assesses whether two groups differ. It is non-parametric and rank-based, making no normality assumption and remaining robust to batch effects across cohorts. With large sample sizes, however, even minor differences become statistically significant, so Cliff's delta complements the test by quantifying how large the difference is [20].

Cliff's  $\delta$  quantifies the probability that a randomly drawn value from one group exceeds one from the other, minus the reverse probability, ranging from  $-1$  to  $+1$ . A positive sign indicates Western enrichment, negative indicates Non-Western enrichment, with magnitude thresholds of negligible ( $<0.147$ ), small ( $<0.33$ ), medium ( $<0.474$ ), and large ( $\geq 0.474$ ) [21].

#### 2.5.4. USA BMI extremes

This part focuses on a targeted exploratory question: Can a predictive microbial signature be extracted when the analysis is restricted to the extremes of the USA BMI distribution? By narrowing the focus to the "tails" of the cohort, Lean (BMI  $< 22$ ) versus Obese (BMI  $> 32$ ), we aim to minimize the noise introduced by intermediate phenotypes (Overweight) and maximize the potential for detecting adiposity-related microbial patterns.

#### 2.5.5. Methodological Strategy and Validation

To address the high-dimensional nature of the metagenomic data and the risk of overfitting, we implemented a robust machine learning pipeline centred on Random Forest (RF) classification. The model was trained using 5-fold Stratified Cross-Validation with balanced class weights to account for the sample size disparity between the lean and obese groups.

A critical component of this analysis is the Label-Permutation Test. Given the lack of clear clustering in the previous UMAP analysis, it is essential to distinguish true biological signals from stochastic noise. By shuffling the class labels 100 times and retraining the model, we generate a null distribution of "random performance." This allows us to statistically validate the model's predictive power; the real cross-validation score must significantly exceed this null distribution to consider the identified microbial features as biologically relevant rather than mathematical artifacts of a high-dimensional feature space.

### 2.5.6. Model performance and statistical validation

The Random Forest model achieved a Balanced Accuracy of 0.570 ( $\pm 0.196$ ) and a Macro-F1 Score of 0.561. While these values are not high in absolute terms, they must be interpreted within the context of the high-dimensional and sparse nature of microbiome data. The variance across folds (ranging from 0.50 to 0.75) reflects the inherent heterogeneity of the small USA extreme-BMI cohort.

To determine if this predictive performance represents a true biological signal or a stochastic artifact, we conducted a label-permutation test ( $n = 100$ ).

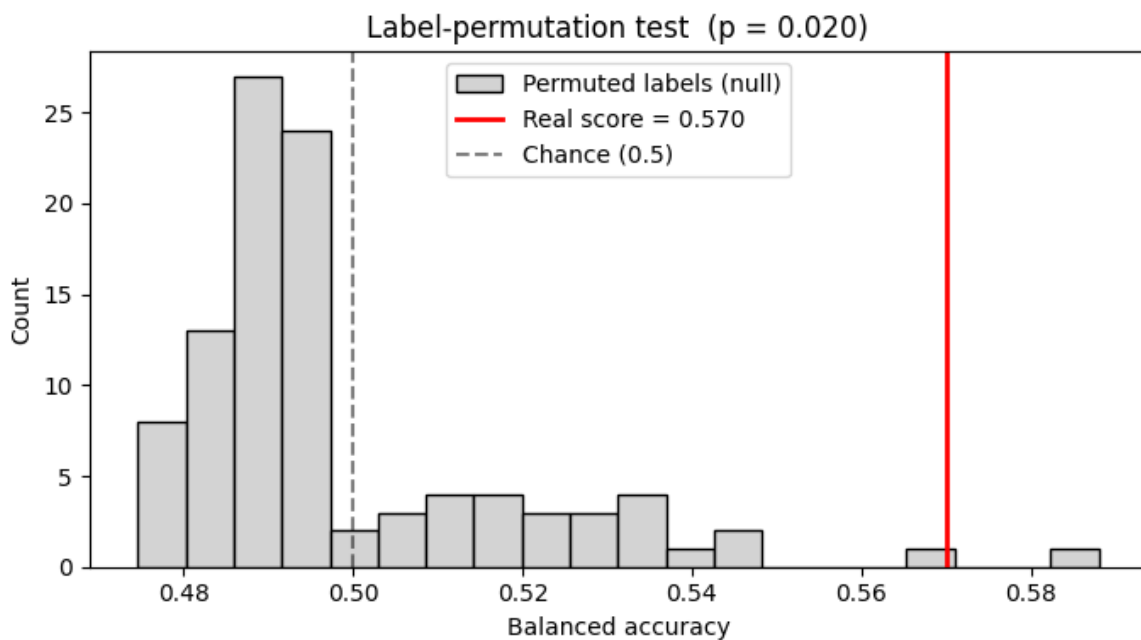


Figure 11: Null distribution of cross-validated balanced accuracy obtained from 100 random label permutations of the Lean vs. Obese Random Forest model (grey histogram). The dashed line marks the chance baseline (0.5) and the red solid line marks the observed model score (0.570,  $p = 0.02$ )

With a cross-validation score of 0.570 ( $p = 0.02$ ), the model performs significantly better than the random null mean of 0.498. This confirms that while microbial differences between lean and obese individuals are too subtle for visual clustering, a Random Forest can still detect a complex signature that distinguishes them. Since this signal is statistically valid and above the noise floor, we can now examine the specific microbial taxa driving these predictions.

## 3. Results

### 3.1. Microbial Signatures of Western vs. Non-Western samples

The aim of this analysis was to characterise the microbial differences between Western and Non-Western cohorts. As described in the *Materials and Methodology* section, feature attributions were extracted from both classifiers to identify the microbial species most strongly associated with each lifestyle. However, the two models produced partially divergent rankings, as shown in Figures 12 and 13.

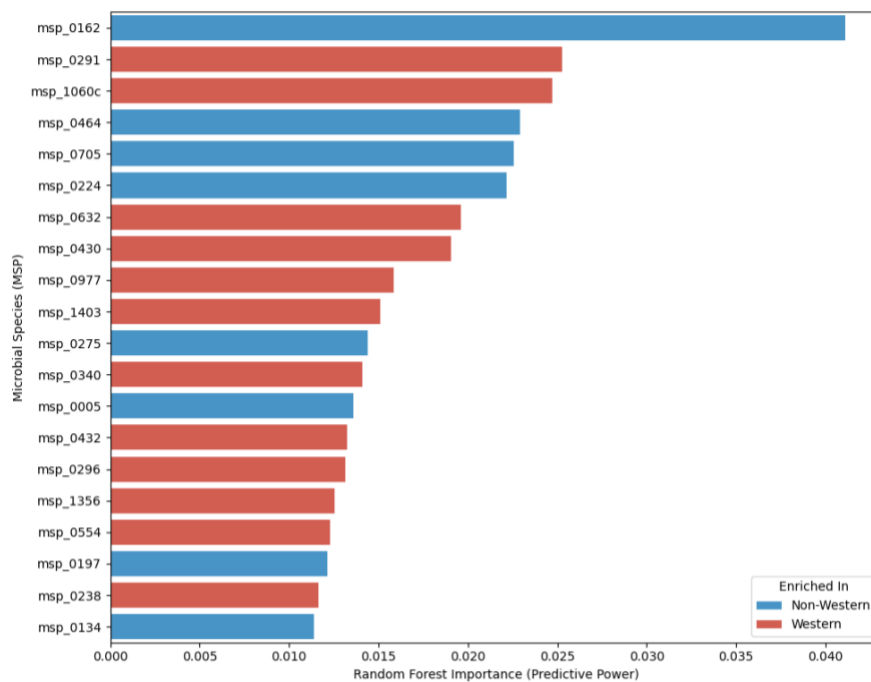


Figure 12: Top 20 microbial predictors of Western vs. Non-Western lifestyle from the Random Forest model, ranked by feature importance. Bar colour indicates the direction of enrichment based on CLR-transformed group means (Western vs. Non-Western)

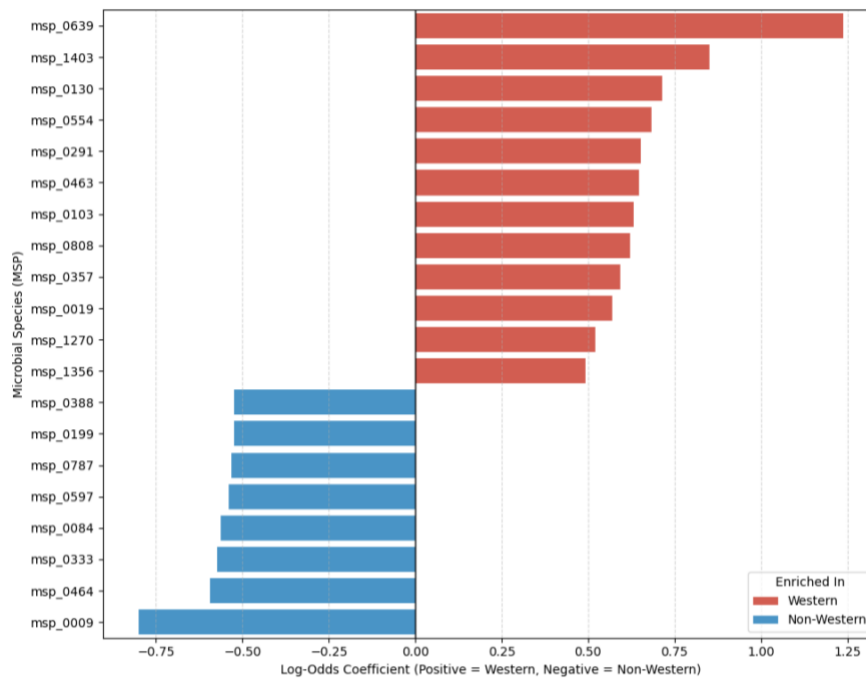


Figure 13: Top 20 microbial predictors of Western vs. Non-Western lifestyle from the L1-regularised Logistic Regression model. Bars show the log-odds coefficient of each MSP, with positive values indicating enrichment in Western samples (red) and negative values indicating enrichment in Non-Western samples (blue)

Across the top 20 features from each model, five MSPs were shared by both classifiers, all with concordant directionality. The 5 microbiomes are listed in the Table 2.

Table 2 Microbial feature importance from Random Forest and logistic regression. The table details five key taxa driving classification between Western and Non-Western cohorts based on effect size ( $\delta$ ), strength, and adjusted p-values

| MSP Number | Bacteria Name                |
|------------|------------------------------|
| msp_0291   | <i>Ruminococcaceae</i>       |
| msp_0464   | <i>Eubacterium ramulus</i>   |
| msp_1356   | <i>Anaerostipes</i>          |
| msp_1403   | <i>Anaeromassilibacillus</i> |
| msp_0554   | <i>Blautia</i>               |

The 5 microbes were statistically validated. Results are presented in Table 3.

Table 3 Four microbes (*Ruminococcaceae*, *Anaerostipes*, *Anaeromassilibacillus*, and *Blautia*) show a statistically significant enrichment in the Western group with medium to large effect sizes. Conversely, only *Eubacterium ramulus* is enriched in Non-Western

| Microbe                      | Enriched in | Effect size ( $\delta$ ) | Strength | p-value (adjusted) |
|------------------------------|-------------|--------------------------|----------|--------------------|
| <i>Ruminococcaceae</i>       | Western     | 0.56                     | Large    | 1.26e-161          |
| <i>Eubacterium ramulus</i>   | Non-Western | -0.55                    | Large    | 1.06e-157          |
| <i>Anaerostipes</i>          | Western     | 0.43                     | Medium   | 1.09e-95           |
| <i>Anaeromassilibacillus</i> | Western     | 0.42                     | Medium   | 3.67e-91           |
| <i>Blautia</i>               | Western     | 0.36                     | Medium   | 8.18e-68           |

### 3.1.1. Western Diet and Systemic Inflammation: Literature Context

A western-style diet promotes chronic systemic inflammation through the gut microbiota. Research shows that this dietary pattern shifts microbial composition and weakens the gut barrier, contributing to systemic inflammation linked to obesity, type 2 diabetes, and cardiovascular disease [22].

#### *Ruminococcaceae*

*Ruminococcaceae* are anaerobic bacteria that ferment complex dietary fibres in the human gut. As primary degraders of resistant starch, they possess specialised enzymatic machinery for substrates that most other gut microbes cannot process. Their fermentation produces short-chain fatty acids (SCFAs), which regulate inflammatory and neuroimmune pathways. High abundance characterises metabolically healthy profiles and is associated with reduced systemic inflammation. Their depletion has been linked to inflammatory bowel disease, cardiovascular disease, and type 2 diabetes, as reduced butyrate production weakens the gut barrier and the body's anti-inflammatory capacity [23], [24].

#### *Eubacterium ramulus*

*Eubacterium ramulus* is an anaerobic gut bacterium that metabolises dietary flavonoids, contributing to the bioavailability of flavonoid-derived metabolites in the host. Despite this seemingly beneficial function, elevated levels characterise an "unhealthy" gut profile. *E. ramulus* is frequently co-observed with taxa such as *Streptococcus* and *Anaeromassilibacillus*, a pattern typically associated with low-fibre, processed diets. Its enrichment has been linked to systemic inflammation and increased risk of cardiovascular disease and type 2 diabetes, suggesting it reflects broader dietary and microbial imbalances rather than a direct pathogenic role [24].

#### *Anaerostipes*

*Anaerostipes* is a common anaerobic genus, typically representing 2–7% of the gut microbial community. Its primary role is butyrate production, which supports gut barrier integrity and modulates immune function. Higher abundances characterise metabolically healthy profiles and are associated with lower body fat, reduced triglycerides, and elevated HDL ("good") cholesterol. Depletion has been linked to inflammatory bowel disease, type 2 diabetes, and increased cardiovascular risk. One clinically relevant caveat: some strains can metabolise the

chemotherapeutic agent 5-fluorouracil (5-FU), potentially reducing its efficacy in colorectal cancer treatment [24], [25], [26].

### *Anaeromassilibacillus*

*Anaeromassilibacillus* is an anaerobic genus typically observed in individuals with low dietary fibre intake. Unlike fibre-fermenting taxa such as *Ruminococcaceae*, it does not produce beneficial short-chain fatty acids and thrives in gut environments with reduced microbial fermentation capacity. Elevated abundances have been linked to impaired intestinal barrier integrity, which may permit translocation of pro-inflammatory compounds into the bloodstream and contribute to systemic low-grade inflammation. Its enrichment is associated with increased risk of cardiovascular disease and type 2 diabetes, marking it as a feature of an inflammation-prone, low-fibre gut profile [24].

### *Blautia*

*Blautia* is a common anaerobic genus with documented anti-inflammatory and probiotic properties, contributing to metabolic regulation and gut barrier integrity. Higher abundances characterise metabolically healthy profiles and have been associated with lower body fat, reduced triglycerides, lower medication use, and a decreased five-year risk of cardiovascular disease. Together with *Agathobacter* and *Christensenellaceae*, *Blautia* is considered a hallmark of a diverse, stable gut microbiome supporting metabolic health across the life course. Its depletion has been linked to obesity, type 2 diabetes, inflammatory bowel disease, liver cirrhosis, and colorectal cancer [24], [27].

## 3.2. Microbial Signatures of BMI (USA cohort)

The goal of this analysis was to identify which specific microbial species drive the differences between Lean (BMI < 22) and Obese (BMI > 32) individuals in the USA cohort. While unsupervised methods (UMAP) did not show clear clusters, the supervised Random Forest model successfully extracted a statistically valid signature.

### 3.2.1. Feature extraction

Random Forest classifiers give out feature importance ranking based on their mean decrease in impurity (MDI), also known as Gini importance. MDI measures how much each microbial feature (MSP) contributes to “purifying” the data during the decision-making process of the model. Every time a tree splits based on a specific bacterium, the impurity (uncertainty) of the classification decreases. MDI is the average of these decreases across all trees in the forest.

A higher MDI indicates that the bacterium is a more powerful predictor for distinguishing between lean and obese phenotypes.

### 3.2.2. Top Drivers

By analysing the CLR-transformed mean abundances, we determined whether these high-importance features were enriched in the lean or obese group.

*Table 4 Top microbial species by Random Forest MDI importance. The table lists five key species driving classification between obese and lean phenotypes. *Blautia* sp. and *[Clostridium]* sp. exhibit the highest feature importance and are enriched in the obese group*

| Species (canonical)                         | MDI Importance | Enrichment |
|---------------------------------------------|----------------|------------|
| <i>Blautia</i> sp. (msp_0142)               | 0.0243         | Obese      |
| <i>[Clostridium]</i> sp. CAG:169 (msp_0357) | 0.0178         | Obese      |
| <i>Alistipes shahii</i> (msp_0196)          | 0.0131         | Lean       |
| <i>Bacteroides uniformis</i> (msp_0046)     | 0.0120         | Lean       |
| <i>Alistipes onderdonkii</i> (msp_0032)     | 0.0094         | Lean       |

As shown in Table 4, the model identified a distinct taxonomic split between the two phenotypes. The strongest overall predictor, *Blautia* sp. SG772 (MDI: 0.0243), was enriched in the obese cohort alongside *[Clostridium]* sp. CAG:169.

The lean phenotype, by contrast, was primarily driven by species of the *Bacteroidetes* phylum, notably two *Alistipes* species (*A. shahii* and *A. onderdonkii*) and *Bacteroides uniformis*. This clear biological division suggests that while overall community shifts are difficult to detect visually, specific taxa play outsized roles in the metabolic environments associated with extreme BMI profiles.

*Blautia* sp. SG772 and *[Clostridium]* sp. CAG:169 have not yet been formally named. Their genomes were identified through metagenomic binning from complex sequencing data, meaning each organism is defined only by its genetic signature in a reference catalogue rather than by a formally described, cultured isolate.

### 3.2.3. Literature validation vs. Study Findings

To contextualize the microbial signatures identified in the USA cohort, we mapped the top-ranking features (MSPs) from the Random Forest model against established clinical literature. This synthesis evaluates whether the identified species are known biomarkers for metabolic health or systemic dysfunction.

### 3.2.4. Global studies

Current research indicates that the gut microbiome is a primary mediator of metabolic health, influencing everything from adipose tissue inflammation to the gut-brain axis.

*Table 5 Literature validation of key microbial taxa and phenotypic enrichment. The table maps identified taxa to documented clinical links and evaluates literature alignment. Bacteroides uniformis, Alistipes shahii, and [Clostridium] sp. show full consensus with literature*

| MSP ID   | Taxon                            | Enrichment | Clinical Link (Literature)      | Alignment     |
|----------|----------------------------------|------------|---------------------------------|---------------|
| msp_0046 | <i>Bacteroides uniformis</i>     | Lean       | Anti-obesity / Metabolic Health | Consensus     |
| msp_0196 | <i>Alistipes shahii</i>          | Lean       | Gut Barrier Protection          | Consensus     |
| msp_0052 | <i>Blautia sp.</i>               | Unclear    | -                               | Partial       |
| msp_0357 | <i>[Clostridium] sp. CAG:169</i> | Obese      | Fatty Liver / Metabolic Stress  | Consensus     |
| msp_0032 | <i>Alistipes onderdonkii</i>     | Lean       | Depression / High BMI Marker    | Contradiction |

#### *Bacteroides uniformis*

Evidence suggests this strain protects against diet-induced obesity by improving glucose tolerance, lowering cholesterol, and boosting anti-inflammatory butyrate production, acting as a metabolic stabilizer [28].

#### *Alistipes shahii*

This species is a marker for a healthy gut barrier and its integrity. It has been shown to reduce mucosal inflammation and shift the immune response toward an anti-inflammatory profile [29].

#### *Alistipes onderdonkii*

Linked to clinical depression and metabolic imbalances, this species is known for breaking down tryptophan, a precursor for serotonin, potentially linking gut dysbiosis to neuropsychiatric symptoms [30].

#### *Clostridium sp. CAG:169*

A robust marker for non-alcoholic fatty liver disease (NAFLD) and obesity-related hepatic stress; its abundance is typically inversely correlated with diet quality [31].

### *Blautia* sp. SG772

The role of the *Blautia* genus in obesity is complex and highly species-specific. While its overall impact can be ambiguous, specific beneficial species (such as *B. wexlerae*, *B. hansenii*, and *B. luti*) demonstrate strong probiotic potential by producing short-chain fatty acids (SCFAs), improving insulin sensitivity, reducing visceral fat accumulation, and supporting intestinal barrier integrity and anti-inflammatory responses [32] .

## 4. Discussion

### 4.1. Western vs. Non-western

The five microbial species agreed on by both classifiers all reached high statistical significance and medium-to-large effect sizes. This convergence between two methodologically different models strengthens confidence that these signals reflect genuine biological differences rather than single-method artefacts.

The findings partially support the hypothesis that Western diets shift the microbiome toward a low-fibre, inflammation-prone configuration [22]. The Non-Western enrichment of *Ruminococcaceae* is consistent with higher fibre intake in traditional diets, supporting butyrate production and gut barrier integrity. The Western enrichment of *Eubacterium ramulus* and *Anaeromassilibacillus* fits the same picture, as both are characteristic of low-fibre, processed diets and associated with weakened barrier integrity and elevated cardiometabolic risk.

More surprising is the higher abundance of *Anaerostipes* and *Blautia* in Western guts, as both genera are commonly linked to metabolic health. This likely reflects that the Western/Non-Western distinction captures many lifestyle factors beyond diet. Moreover, gut health is best evaluated through overall microbial diversity and community composition rather than individual taxa, so a microbe that appears beneficial in isolation can be enriched in contexts that are not, overall, healthier.

### 4.2. BMI

The alignment between our findings and existing literature for *B. uniformis*, *A. shahii*, and *Clostridium* sp. CAG:169 provides strong biological validation for our supervised learning pipeline. It confirms that even in a high-dimensional dataset where visual clustering (UMAP) was absent, the Random Forest model successfully identified a signature of "metabolic stabilisers" and "metabolic stressors".

In the USA cohort, the lean phenotype (BMI < 22) is characterized by barrier-protecting commensals and metabolic stabilizers.

*B. uniformis* and *A. shahii* are enriched in this group, consistent with their roles in improving glucose tolerance and maintaining gut barrier integrity. The obese phenotype (BMI > 32), in contrast, is marked by species associated with systemic inflammation and metabolic dysfunction. *Clostridium* sp. CAG:169 acts as a marker for obesity-related stress and is specifically linked to non-alcoholic fatty liver disease.

A notable discrepancy was found with *A. onderdonkii*, which was enriched in our lean cohort despite being identified in the literature as a marker of high BMI and depression. This underscores that the gut-adiposity axis is highly complex and that the role of specific species may be context-dependent, shaped by the geographical and dietary characteristics of the USA cohort. The finding highlights the need for regional, cohort-specific analyses in microbiome research. This complexity is further exemplified by *Blautia*, which shows a highly species-specific relationship with obesity and can act as either a metabolic stabiliser or stressor depending on the strain. Probiotic species such as *B. wexlerae*, *B. hansenii*, and *B. luti* align with lean, healthy phenotypes, while others correlate positively with obesity markers, for example *B. obeum* with increased visceral fat [32].

### 4.3. Limitations

Several limitations should be noted. The cross-sectional design captures associations rather than causal relationships and cannot reveal how the microbiome develops over time or responds to changes in diet, lifestyle, or body weight. The binary Western/Non-Western classification compresses considerable within-group diversity in diet, medication, hygiene, and host genetics into a single label, and BMI serves only as a coarse proxy for metabolic health. The cohort is also geographically imbalanced, with Western populations strongly overrepresented, which may bias both ordination and classification toward the dominant subgroups. In addition, the MSP-level resolution may obscure biologically relevant differences between strains of the same species. This is illustrated by the genus *Blautia*, in which different strains have been linked to opposing effects on host metabolic health.

### 4.4. Outlook

Several directions could strengthen and extend the present findings. Addressing the limitations outlined above, through longitudinal designs, more balanced cohorts, and strain-level resolution, would strengthen the generalisability and biological interpretability of these findings. Integrating functional readouts such as metabolomics or short-chain fatty acid quantification, together with detailed dietary and clinical metadata, would make it possible to test whether the compositional differences observed here truly translate into the inflammatory and metabolic effects described in the literature. Finally, since the present analyses were restricted to samples from healthy individuals, extending the framework to include cohorts with relevant diseases (such as inflammatory bowel disease, type 2 diabetes, or cardiovascular conditions) would allow the identified microbial signatures to be evaluated in a clinical context and clarify whether they retain discriminative power across health and disease states.

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# Appendix

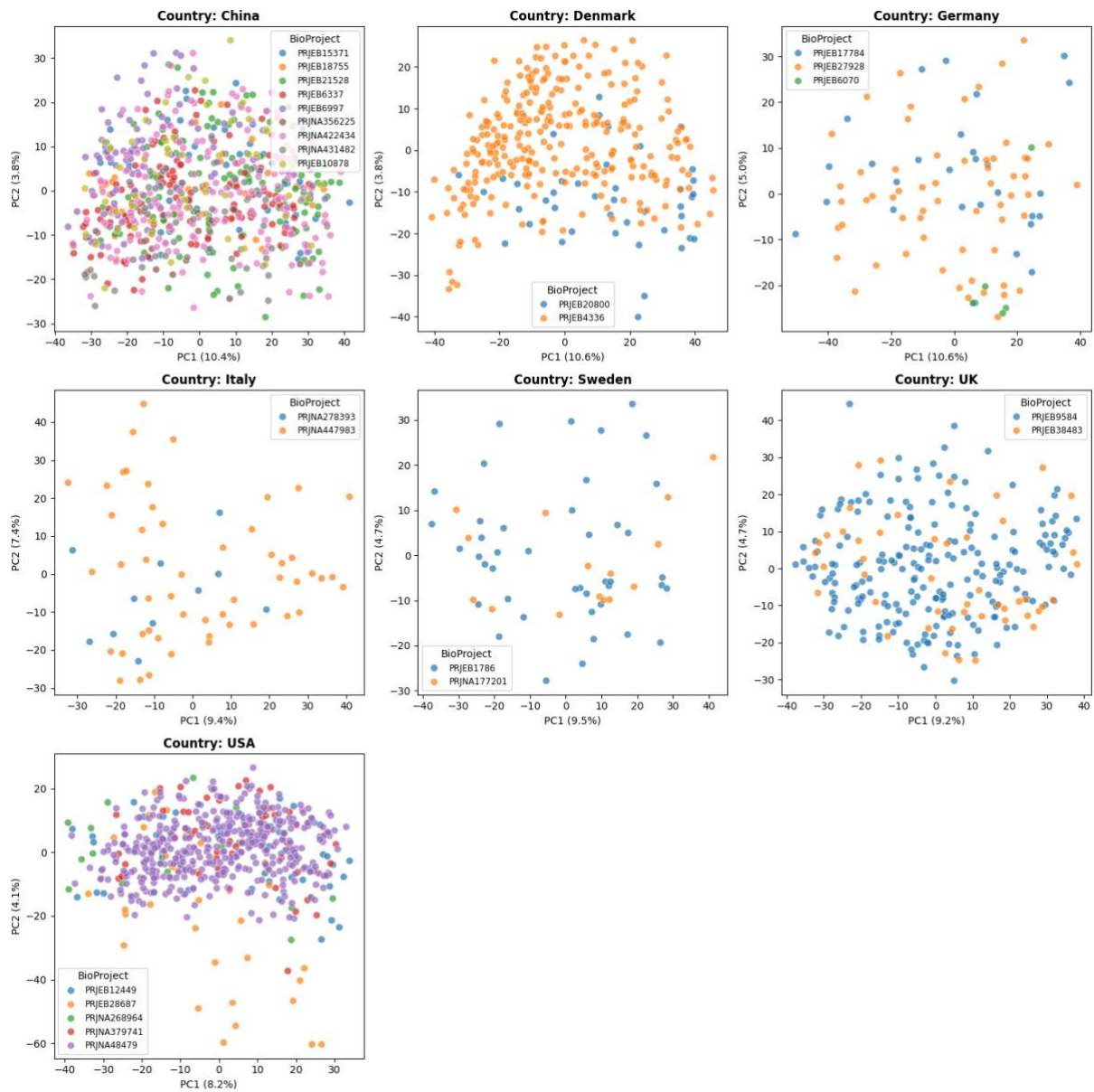
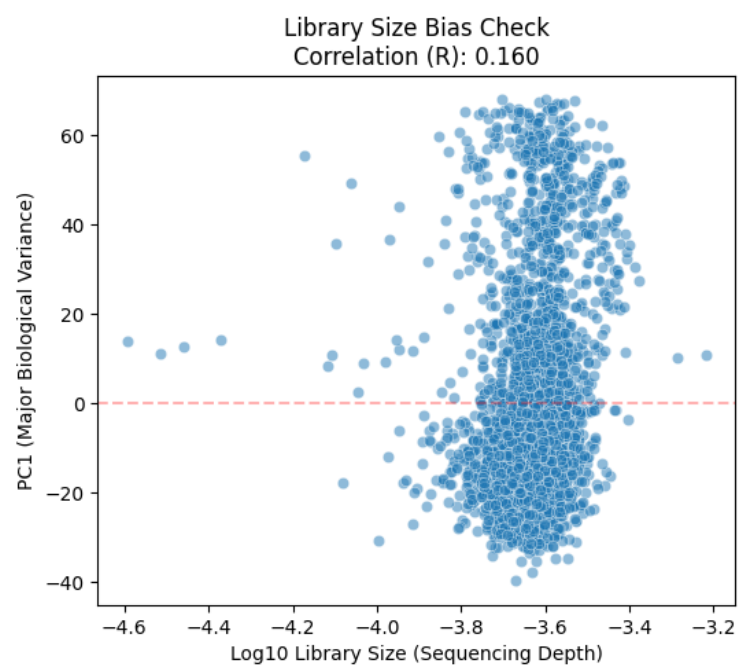


Figure 14 PCA of microbiome profiles across countries with multiple datasets to check for Batch-Effect



*Figure 15 Library-size assessment, correlation between first principal component and library size (with correlation  $R = 0.16$ )*

# Declaration on the use of AI-Based Tools

We hereby declare that generative Artificial Intelligence (AI) tools were utilised as auxiliary aids during the preparation of this project assignment (PA1). These tools were deployed responsibly, primarily to enhance workflow efficiency and provide editorial support. The contribution of generative AI tools to the creative output of this assignment is documented below and recognisable to third parties. All AI-generated suggestions were critically evaluated, verified, and supplemented by our own formulations and analyses.

## Method Reference

AI tools were used in a supportive capacity across three areas: assistance with debugging and optimisation of Python code, linguistic proofreading and stylistic refinement of the English manuscript to maintain an appropriate academic register and meet length requirements; and literature discovery and synthesis of initial overviews of current scientific consensus. At no point were AI-generated outputs adopted without critical verification.

## List of AI Tools Used

Anthropic. (2026). *Claude* (version Sonnet 4.6). <https://claude.ai>: Linguistic proofreading and stylistic refinement; debugging of Python code

Google. (2026). *Gemini* (version 3.5 Flash). <https://gemini.google.com>: Debugging of Python code; optimisation of data filtering scripts; generation of code snippets for statistical evaluations; stylistic refinement of the English manuscript

Microsoft. (2026). *Copilot* (version as accessed). <https://copilot.microsoft.com>: Assistance with debugging Python code

DeepL. (2026). *DeepL Translator*. <https://www.deepl.com/translator>: Machine translation of the abstract into German

Google. (2026). *NotebookLM*. <https://notebooklm.google.com>: Literature discovery and synthesis of initial overviews of microbiome-host associations

Consensus. (2026). *Consensus AI* (version as accessed). <https://consensus.app>: Locating relevant peer-reviewed studies and synthesising scientific consensus